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Engineering and Technology

TECH INTELLINA 2024

PID-1.7

Stress-Sense-Pedal

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Table of contents

01

Abstract

02

Objective

03

Introduction

04

Social Impact

05

Solution Approach

06

Model Demonstration

07

Cost Involvement

08

Conclusion

09

References



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01

Abstract

Abstract

Introducing our project , the stress sense pedal. This project was made in mind the regular **high levels of stress** that people experience.

Our main aim was to measure the stress level of the user using their physical interactions and reactions and put those actions to use as a way of **measuring their stress level**.

We aimed to help people the amount of stress that they have to experience on a daily basis and help them to realise the **detrimental effects** it has on their mindset and body.

This device has made significant strides in assessing stress levels and providing insights into both **mental well-being** and **physical health**.



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02

Objective



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Objective

Stress Measurement

Measure the amount of stress a person has to go through on an everyday basis

Stress-level information

Letting them know about the jobs/ experiences that gives them extra stress.

Mental relaxation awareness

Informing them about the multiple detrimental effects it may have on their mind, body and their daily life.



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03

Introduction

Introduction

- The **Stress Sense Pedal** is an innovative device designed to monitor stress levels by analyzing leg movement patterns.
- By measuring the **frequency of leg shaking or tapping**, the pedal provides real-time insights into an individual's stress levels.
- The device is made in such a way that the device captures the **number of times** a person taps their feet.
- It then **collects the data** and **analyzes** it. Based on the leg shaking patterns, the pedal calculates a stress score. Higher leg shaking frequency may indicate elevated stress levels. If stress is detected then it alerts the user about it .

STRESS SENSE PEDAL

CREATED BY...

SHYANTAN KUNDU, 2nd yr CSE
SOHAM SARKAR, 2nd yr ECE

Working Of the Project...

- STRESS-SENSE PEDAL MEASURES STRESS LEVELS THROUGH LEG-SHAKING, MAPPING THEM TO AN ANALOG SIGNAL.
- MICROCONTROLLER PROCESSES DATA, DISPLAYING STRESS LEVELS ON AN OLED SCREEN FOR USER CONVENIENCE.
- MODULAR DESIGN ALLOWS FOR FLEXIBILITY AND EASY MAINTENANCE, ENSURING LONG-TERM FUNCTIONALITY.

Need for the InVeNtioN

The need for the StressSense Pedal project lies in providing a non-intrusive, real-time stress monitoring solution.

Its uniqueness lies in its ability to measure stress levels through leg-shaking, offering a novel approach to stress management.

This innovation offers a discreet and effective way to track and alleviate stress.

Our Motto

"Experience stress,
go with StressSense."

04

Social Impact

04

Social Impact

- This projects at helping people realize the jobs and works they do in order to experience stress and then help them to lessen the stress causing factor in their life.
- Out objective is to help gain better lives free of stress and helping them adopt a better while seamlessly transitioning from their current lifestyle.
- It also helps in releasing stress because it allows the person the option of physical movement all the while sitting in a single place.



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05

Solution Approach

05

Solution Approach

- By measuring the amount of stress one has to cope with during work or in a specific work environment, necessary changes can be made to the work schedule or the workspace so that the level of stress is reduced over their mind.
- Thus sky rocketing their **productivity** all the while taking ample care of their mental health status which is often disregarded in this fast forwarded life.
- Our project also provides stress relief by providing the person allowance for physical movement while sitting in a place.



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06

Model Demonstration



06

Model Demonstration



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07

Cost Involvement

07

Cost Involvement

Materials	Cost
Wood	200
micro controller	350
Piezo-electric plates	30
sensors	80
OLED screen	160
9v Battery	20
Battery Case	20
Switch	10
Wires	40
Breadboard	80
Metal sheet	50
Total Cost	800



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08

Conclusion

08

Conclusion

- The Stress Sense Pedal uses leg shaking frequency to monitor stress levels in real-time.
- It helps individuals become more aware of their stress patterns.
- Users can take proactive steps for better mental and physical well-being based on the insights provided.
- Employers can use the pedal to promote workplace health and stress management programs.
- The device empowers users in their journey toward a healthier and more balanced life by increasing stress awareness.



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Project ID: PID-1.7 Stressaway

Do you have any questions?

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